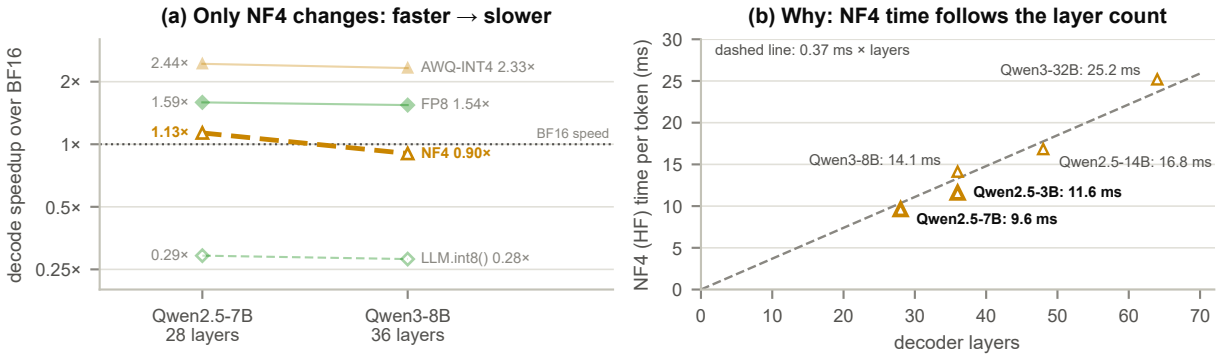
(a) Only NF4 changes: faster → slower
decode speedup over BF16
2×
1×
0.5×
0.25×
2.44×
AWQ-INT4 2.33×
1.59×
FP8 1.54×
1.13×
NF4 0.90×
BF16 speed
0.29×
LLM.int8() 0.28×
Qwen2.5-7B
28 layers
Qwen3-8B
36 layers
(b) Why: NF4 time follows the layer count
NF4 (HF) time per token (ms)
dashed line: 0.37 ms × layers
Qwen3-32B: 25.2 ms
Qwen3-8B: 14.1 ms
Qwen2.5-14B: 16.8 ms
Qwen2.5-3B: 11.6 ms
Qwen2.5-7B: 9.6 ms
0
5
10
15
20
25
30
0
10
20
30
40
50
60
70
decoder layers